

The Joy of Sketch: A Hands-on Introductory Course on Sketching in HCI and UX within Research, Practice, and Education

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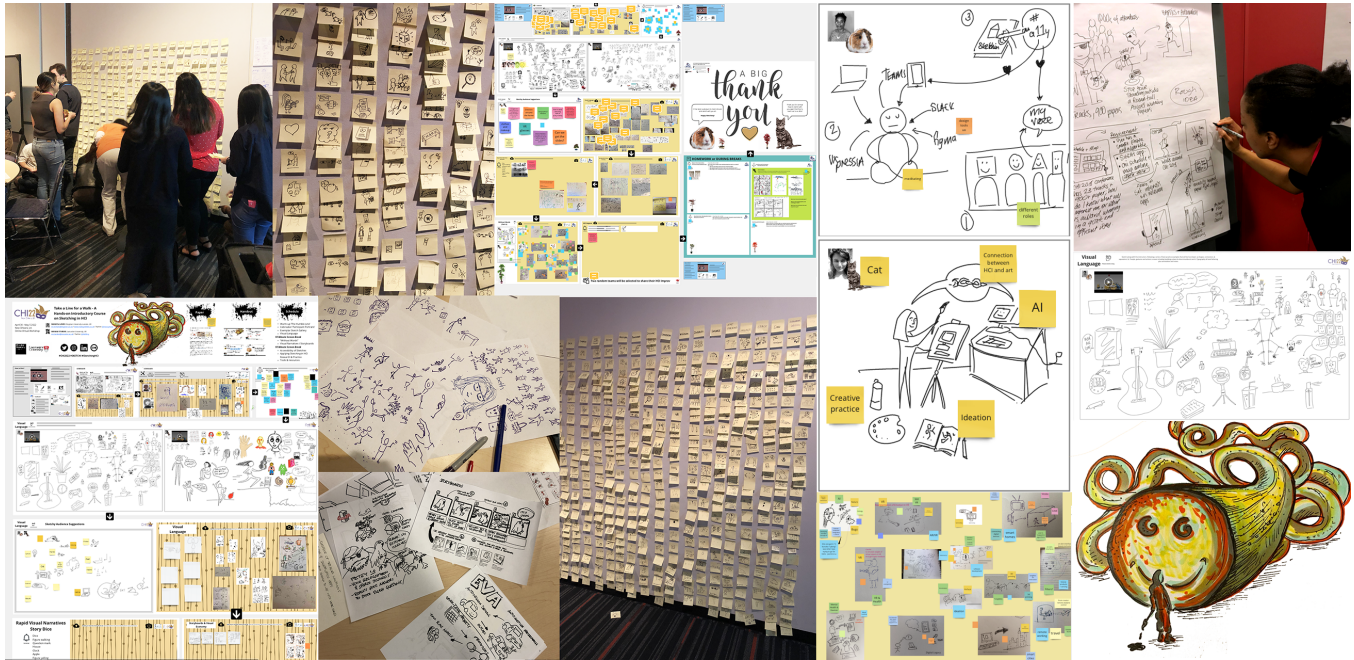


Figure 1: Five years of Sketching in HCI: A collage of courses at CHI from 2018-2022, including in-person and online.

ABSTRACT

Sketching is a powerful tool for communication, expression, and interrogation – and yet it is also a joyful method that connects people and makes research and education more engaging. Whether hands-on with pen and paper or on a tablet, the manual action of creating an image to tell the story of research is a valuable addition to any student or professional's skill set in Human-Computer Interaction and User Experience Design. We will take you on a complete journey in sketching, from the basics of mark-making right through to objects, interfaces, and interactions with people.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI).**

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CHI EA '23, April 23–28, 2023, Hamburg, Germany

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ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3574187>

KEYWORDS

sketching, drawing, comics, storyboards, UX

ACM Reference Format:

Makayla Lewis and Miriam Sturdee. 2023. The Joy of Sketch: A Hands-on Introductory Course on Sketching in HCI and UX within Research, Practice, and Education. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3544549.3574187>

1 BENEFITS

Of all the techniques we use in our day-to-day practice for HCI and UX, sketching is the most timeless, and still offers new routes into research – whether as a method, tool, or the subject of interrogation itself. Mark making is fundamentally human activity and has its roots in the formation of language. Sketching allows us a window into the mind and offers a collaborative space where we can reflect and iterate. Sketching is a fast, easy and resource-light method of communicating, iterating, prototyping and illustrating our ideas, concepts and devices, surfaces and spaces. Further, sketching can also benefit those who work in the programming spaces to make sense of thoughts and outputs. It can be a solitary and subjective

activity to work through a problem, or a piece of collaborative research which invites others to add, subtract or edit – there are no boundaries. We intend this course to help participants to get over the ‘fear of the blank page’ and will teach easy, accessible methods of image-making via sketching for HCI and UX. Participants will come away from the sessions with the confidence to create useful sketches with which they can enhance their own research, industry practice or teaching. The ran the last two years’ courses online using Miro as a collaborative tool (CHI 2022 Miro Board miro.com/app/board/uXjVO6FZQx8=/?share_link_id=375118978821), but for CHI 2023 we are delighted to return in person to host our Sketching in HCI and UX course. We will also be offering a remote session for those of our community who are unable to join us in Germany. This course links into our upcoming textbook on Sketching in HCI which will be published in the Fall of 2023, and attendees will be able to use their new skills in conjunction with the book to pass on their knowledge in practice.

2 INTENDED AUDIENCE

The content of this course is suitable for academics (teachers and researchers), industry leaders and practitioners, students, and early career researchers that have an interest in learning and or improving their sketching skills. Novices, experts, and those with an interest are welcome to attend. The instructors intend to recruit participants through relevant mailing lists, and by reaching out to researchers, practitioners, and students that have previously shown an interest in sketching in HCI and UX. To encourage international community building, social media will be used e.g., Twitter and Instagram using hashtags #HCI #Sketching #CHI2022 as well as the course hashtag #SketCHI. A course website is available at: www.sketchhci.wordpress.com/.

3 PREREQUISITES

No prerequisites are required, although attendees should have an interest sketching.

4 CONTENT

Sketching is often overlooked in many disciplines, this includes HCI and UX, and it is frequently demoted as a ‘soft’ skill or creative pursuit that is kept behind the scenes. However, research has shown sketching supports planning, ideation [6], prototyping [2], collaboration [8], documentation [25], exploration, and the discovery of new and complex spaces [23]. This introductory course intends to share the ‘joy of sketching’ using hands-on practical activities. We will support participants in exploring, learning, sharing, and discussing the role of sketching in HCI and UX. It is hoped this will inspire greater adoption of sketching into their everyday research and practice.

5 PRACTICAL WORK

Course participants will be guided through selected sketching techniques and strategies (see examples, Figure 1). These techniques will be based on well-established sketching, interaction design and computer science material, e.g. [9, 10, 19], but will also include additional techniques and examples. The activities have been especially designed for SIGCHI community based on observations

and feedback from previous CHI sketching courses by the authors [14, 16–18]. The course will be structured as follows:

1. Warm-up ‘The Humble Line’: Activity #1: Participants will be asked to embrace their ‘younger selves’ by mark-making (scribbling), the activity purpose is to let go of perfection.

2. Icebreaker ‘Participant Portraits’: Activity #2: Participants will be asked to work in pairs to draw each other. They will then be asked to give their drawing to the person and ask: what is your name? where do you work and what is your role? Why have you joined today’s Sketching in HCI course?

3. Exemplar Sketch Gallery: Exemplar presentation and discussion outlining visual thinking, sketching, and sketchnotes with examples from HCI, UX, interaction design and computer science followed by a question and answer to establish participants key motivations and goals.

4. Visual Language: Participants will sketch-along with the instructors, following a series of best practice examples that will be live drawn and digitally projected for immediacy: Activity #3: Shapes, connectors, & separators. Activity #4: People, gestures, and actions, and Show & Tell. Activity #5: Scenes including buildings, place (indoors/outdoors), and Show & Tell. Activity #6: Icon dictionary, participants will work together to rapidly build a visual dictionary of objects and concepts present in HCI. Activity #7: Typography & hand lettering, participants will explore the role of annotation and notes in sketches by hand-lettering using instructors’ examples (worksheets) and Show & Tell. Followed by Activity #8: Sketchy audience suggestions, a rapid sketching session whereby participants ask the instructors to draw an action, screen, object, or concept not previously demonstrated. Followed by an exemplar presentation and discussion outlining the role of color & shading (color theory) in sketching.

5. Sketching in HCI Research and Practice: Exemplar presentation and discussion outlining visual thinking and sketching from HCI, interaction design and computer science. Activity #9 Participants will be asked to sketch their research area or industry practice (e.g., a recent project) without the use of text or verbalization. Sketches will be placed in a ‘sketch gallery’, an easy to access wall or large table within the course room. Activity #10: Instructors and participants will produce a visual mind map exploring significance, benefits, and pitfalls of sketching in HCI and how participants may apply sketching into their everyday work and research practice. Using post-it notes participants will be asked to identify the field of study and key insights of each sketch, each post-it will be stuck next to each sketch. The purpose of this activity is to get to know course peers and to provide constructive feedback on narrative depiction.

6. Visual Narratives: Storyboards & comics exemplar presentation followed by Activity #11: An instructor led group discussion about the use of comics and scenarios in HCI e.g., data comics [1], storyboards [9, 22] followed by best practice techniques for creating coherent and engaging comics and scenarios. Activity #12: Visual Economy, participants will be asked to draw a scenario/sequence in only three panels, then one panel. The purpose of this is for dissemination and publications where size/length is at a premium.

7. Accessibility of Sketches: Presentation about accessibility of sketches in HCI, best practice and examples, e.g., use of screen readers and the need for text alternatives (alt text) www.w3.org/

WAI/alt/, and how such measures also support search engine optimization, followed by Activity #13: Participants will be asked to return to their Activity #10 outputs and add alt text followed by show and tell with a neighbor ensuring constructive critique is given.

8. Curricula Design & Pedagogy for Sketching within HCI & UX Education: Presentation and demonstration of best practice for in-person, remote, and hybrid sketching sessions with students, fellow researchers and practitioners, authors recent journal paper will be highlighted for further reading [15].

The instructors will ensure feedback is given to each participant throughout the course. Participants will also be provided with crib sheets, practice sheets and post-course activities (further guided sketching practice). Activity #14 Sketch Analysis: participants will be given sketches from existing published work and taken through the methodologies that can be used to generate meaningful data from these visuals. Activity #15 Vignette: participants will be provided with a photograph (with alt text) of a *Rory's Storycubes* roll and asked to create a one-page visual story. Activity #16 HCI improv: participants will be given prompts (fictional problem, user persona, technology and user need) to create a visual storyboard depicting a potential solution. Finally, participants will be invited to join our existing network, *Sketching in HCI* on Slack: a community engaging with sketching research and practice in HCI.

6 INSTRUCTOR BACKGROUND

Makayla Lewis is a lecturer in Computer Science (User Experience Design) at Kingston University London, researching and teaching human factors in business, cybersecurity, smart money, and AI. Makayla is an accomplished visual thinker and sketcher who organizes sketching events and courses and provides visuals for international companies and conferences such as ACM CHI & ISS. Selected/related publications: A Tactile Visual Library to Support User Experience Storytelling [13]; Are You Feeling it? The Use of Comic Strips to Encourage Empathy in Design [11]; I've Got Something to Say: The Use of Animation to Create a Meta-Story About Professional Identity [12]; *Makayla Lewis on The Power of Sketchnoting in UX Design* (an Adobe featured blog) xd.adobe.com/ideas/perspectives/interviews/makayla-lewis-power-sketchnoting-ux-design/. **Miriam Sturdee** is a lecturer at the University of St Andrews, specializing in investigating how sketching and the arts can support the design and development of novel technology. She also has an MFA in Visual Communication from Edinburgh College of Art and is an active illustrator, visual facilitator, and sketch artist. Her research has a particular focus on sketching for shape-changing interfaces, Cybersecurity [26], as well as design fiction and alternative publishing formats [21].

7 RESOURCES

Course notes will be produced before course and shared. Following the course, a visual summary 'sketchnotes', crib sheets, and further practice 'homework' activities will be produced and shared. The Sketching in HCI textbook will be available in the Fall for attendees to refresh their knowledge. Makayla's website www.makaylalewis.co.uk contains HCI sketchnotes, daily sketches and illustrations, and

links to public engagement events e.g., Sketchnote Hangout, SketchnoteLDN, sketching crib sheets and worksheets, and an Adobe XD sketching feature. Makayla and Miriam also have multiple publications in the area of sketching and arts in HCI, including our ACM Interactions features [24, 25] and *Curricula Design & Pedagogy for Sketching within HCI & UX Education* in *Frontiers in Computer Science (HCI)* [15]. Other helpful resources include Sketching User Experiences: The Workbook [9], The Back of the Napkin [19], The Sketchnote Handbook [20], Doodle Revolution [7], Sketching as Design Thinking [10], Visual Thinking [4], Drawing Ideas [3], and My Icon Library [5] which offer a beginners' perspective on different sketching approaches.

8 ACCESSIBILITY

The course will be designed and delivered to be attended by as many people as possible, as such, although we will be hosting the course in person, we will be happy to run a second instance online for those attending remotely. All text and verbal utterances will be clear, simple to understand and in English. All shared imagery will contain Alt Text with appropriate color contrast. All sketching demonstrations will be clearly verbalized, clear to follow, and at an appropriate speed. If the course is conducted online, remote attendees will have access to conference platform closed captioning. A virtual whiteboard platform will be set to allow for zoom in up to 300% without problems, keyboard navigation will also be supported. Opportunities for support, questions and comments will be present throughout the course regardless of in-person or online delivery.

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